MIL-P-48322B (AR)
AMENDMENT 8
16 June 1987
SUPERSEDING
AMENDMENT 7
26 December 1986

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650
AFT WARHEAD SECTION AND STRIPPABLE COATING ASSEMBLY

This amendment forms a part of Military Specification MIL-P-48322B (AR), dated 22 May 1981 and is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble: Change the word "Department" to "Departments".

Title: Delete "Section and Strippable Coating".

1.1: Delete "Section and Strippable Coating".

PAGE 2

2.1: Dwg. 9288037, delete, "Section and Strippable Coating".

Delete: Drawings 9287967 and 9287968 in their entirety.

2.2: Delete "E18 - Test for Rockwell Hardness of Metallic Materials" and substitute the following:

"E10 - Test for the Brinnell Hardness of Metallic Materials. E103 - Rapid Indentation Hardness Testing of Metallic Materials".

PAGE 3

3.4: Delete "(information only)".

3.5.2: Delete "burns" and substitute "burrs".

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Add the following new paragraph:

"3.15 Knurl. The minimum effective depth of the radial grooves in the knurl shall be .012 inches between the first and sixth peaks (See Figure 2)."

Add "Figure 2" as follows:

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Reference Dwg 9287943, "Informational View"

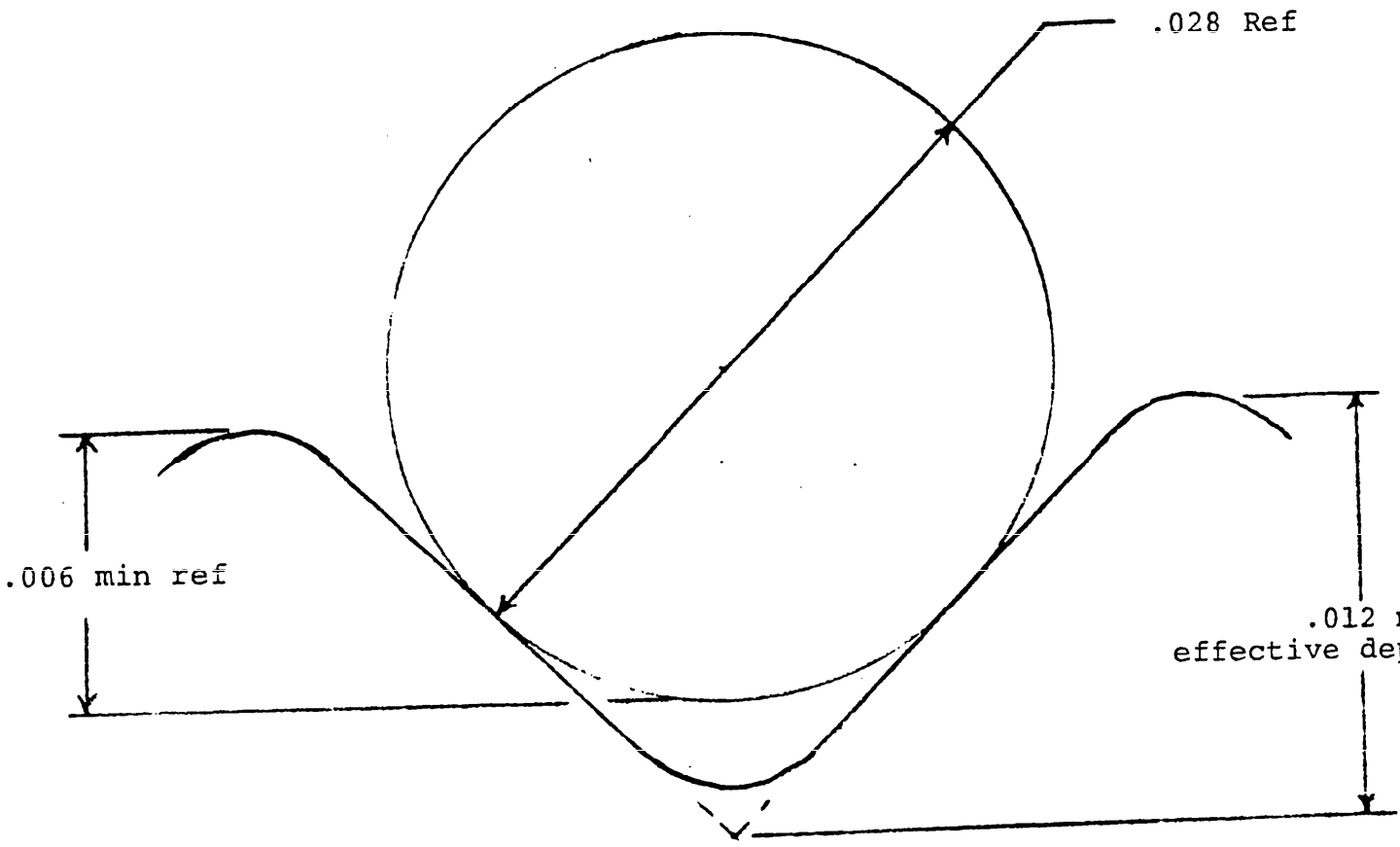


FIGURE 2. Depth of knurl.

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Add the following new paragraph:

"3.15 Knurl. The minimum effective depth of the radial grooves in the knurl shall be .012 inches between the first and sixth peaks (See Figure 2)."

Add "Figure 2" as follows:

PAGE 6

4.3.1: Delete "Section and Strippable Coating" two places.

Delete: "Disk 9288038 10".

PAGE 7

Table I: Delete: "Section and Strippable Coating"

Under "Warhead (Prior to Painting) Metal Parts", delete "Interior Surface Finish (c) 3.9 4.5.6".

Delete: "Base Cavity Radius" and substitute "Base Cavity Radius, if as forged".

Add the following: "Precracked Charpy Test 1 3.4 4.5.3/Appendix".

Add the following: "Precrack Charpy Test 1 3.4 4.5.3/Appendix".

Delete "Disk (dwg. 9288038) 4.4.2.8".

PAGE 8

4.4.1: Delete "and interior surface and internal radius" and substitute "and internal radius, if as forged".

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4.4.2.2: Delete "Critical defect 3 and Note (a) in their entirety.

Add Special defect A as follows:

"Special

A Precracked charpy test 4.4.3.2 3.4 4.5.3/Appendix".

Delete Major defects 103 and 105 in their entirety and substitute the following:

"Major category:

Major 103: Radius of base cavity, if machined 0.40% 3.2 Gage

Major 105: Radius of base cavity, if as forged e 3.2 4.5.5".

4.4.2.3: Delete Major defect 101 in its entirety and substitute the following:

"Major category:

Major 101: Pitch diameter of base thread, min. 0.40% 3.2 Gage".

Add Minor defects 203 and 204 as follows:

"Minor category:

Minor 203: Pitch diameter of base thread, max. 0.65% 3.2 Gage

Minor 204: Major diameter of base thread, max. 0.65% 3.2 Gage".

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4.4.2.4: Add the following new Major defect:

"Major category:

Major 113: Delete "Gage" and substitute "Visual/Comparison Blocks."

Add the following Major defects:

"Major category:

Major 117: Width of knurl, min. 0.40% 3.2 Gage

Major 118: Difference in height of first and eleventh peak of knurl over maximum (measure 3 places, 120° apart) 100% Gage.

Major 119: Depth of radial grooves in knurl, minimum (check three places), (a) 3.15 4.5.14".

Delete Minor defect 212 in its entirety.

Add the following Minor defects:

"Minor category:

Minor 213: Finish of base insulator cavity 0.65% 3.2 Visual

Minor 214: Surface finish 0.65% 3.9 Visual

Minor 215: Evidence of poor workmanship 1.0% 3.14 Visual

Minor 216: Minor diameter of nose thread, min. 0.65% 3.2 Gage"

Add Note (a) as follows:

"(a) See paragraph 4.4.3.11".

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4.4.2.5, Critical defect 2: Change "Visual/4.5.9" to "Visual/Lighting".

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4.4.2.7: Delete Major defect 102 in its entirety and substitute the following:

"Major category:

Major 102: Insulation or adhesive above max. 0.40% 3.2 Gage.

Major defect 103: Delete "100%" and substitute "0.40%".

Add Major Defect 104 as follows:

"104 Adhesive fill around insulator not 360 degrees 100% 3.2 Visual/Ultraviolet Light".

PAGE 17

Delete paragraph 4.4.2.8 in its entirety.

PAGE 18

4.4.2.9: Delete "Section and Strippable Coating".

Delete Minor defects 201, 202 and 203 in their entirety and substitute the following:

"Minor category:

Minor 201: Protective plug missing from forward end 0.65% 3.2 Visual

Minor 202: protective plug, aft end missing 0.65% 3.2 Visual".

PAGE 19

4.4.2.10: Change Dwg. No. from "9288037" to "9280132".

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4.4.3.1.1: In the third sentence change the word "selected" to "selecting".

4.4.3.1.3: In the second sentence change "re-temper" to "reheat treatment".

* Delete 4.4.3.2 in its entirety and substitute the following:

"4.4.3.2 Slow bend precracked charpy test. Beginning with the first heat treat lot produced and continuing until three consecutive heat treat lots have complied with the applicable minimum requirement specified, one final heat treated warhead from each heat treated lot shall be subjected to the precrack charpy test. The warhead selected shall be of the highest average hardness reading from 4.4.3.1.1 and may be the same as used in 4.4.3.1.2. Two specimens shall be tested, with acceptance based on the average fracture toughness. In addition, no individual fracture toughness shall be less than 23 KSI $\sqrt{\text{in.}}$ If one test is invalid, acceptance may be based on the fracture toughness of the valid sample, or another sample from the same warhead may be tested. If either the average or individual fracture toughness is below the minimum allowable, the heat treated lot represented shall be rejected, or subjected to retemper and retest, for all mechanical properties, if approved by the Government. After three consecutive heat treated lots have been accepted, the test shall be discontinued. The test shall be re-instituted, until three consecutive heat treated lots have complied with the applicable requirements, if any of the following occur:

a. Steel is procured from a different producer, or from a substantially different process as determined by the Government.

b. Forgings are procured from a different producer.

c. A substantial change is made in the forging process or configuration, as determined by the Government.

d. A change is made in heat treatment contractors.

e. A substantial change is made in the heat treatment process, as determined by the Government."

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4.4.3.3: Delete in its entirety and substitute the following:

"4.4.3.3 Steel quality. Steel supplied shall comply with MIL-STD-50783, except that macroetch quality shall be per MIL-STD-1459. The quality of the steel as indicated by the macroetch results shall be equal to, or better than A5, B3 and C8 with defects D1, D3, D4, D5, D6, D7 or D8 unacceptable. Manufacturer shall provide a certification that the applicable MIL-S-50783, and macroetch requirements have been met."

4.4.3.5: Delete in its entirety:

4.4.3.7: Delete "3.5.11" and substitute 3.11".

PAGE 22

4.4.3.9: Delete the first sentence and substitute the following

"After qualification in accordance with TTC-490, two salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time."

Add the following new paragraphs:

"4.4.3.11 Depth of knurl.

4.4.3.11.1 Initial sampling. The first pieces from each tool, forming the radial groove, shall be inspected for depth until one part has been accepted.

4.4.3.11.2 Subsequent sampling. If the life of the tool forming the radial grooves is 50 parts or less, the last part formed by each tool shall be inspected. If tool life is greater than 50 parts, an additional inspection of one part from each 50 shall be performed. If the knurl depth is less than .012 inches, production represented by that sample shall be rejected, production shall be stopped and corrective action taken. Sampling for any tool producing a rejected part shall revert to 4.4.3.11.1 if that tool is returned to production."

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4.5.1: Delete in its entirety and substitute the following:

"4.5.1 Mechanical properties-hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining to a sufficient depth, 0.015 inches minimum, to remove scale and get below the decarburization zone. The preparation process to be used shall require government approval. Manual grinding shall not be used. The surface preparation requirement may be eliminated if the following conditions are met:

1. Lab results confirm that current production process produces a decarburized zone of 0.003 inches maximum in the test area. Contractor shall provide samples, or photo-micrographs, of high and low hardness samples to the government for approval. Samples may be the same as for 4.4.3.1.2.

2. A controlled atmosphere is maintained in the austenitizing furnace.

3. Depth of decarburization shall be reverified whenever precrack charpy testing is required, (see 4.4.3.2), but only for the first heat treat lot.

b. Hardness tester periodic verification shall be performed using Brinnell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 10MM Brinnel ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

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e. One reading shall be taken at location X and one at location Y as indicated on Dwg. 9287943."

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4.5.2: Delete the last sentence, "The contractor...DI-T-1906".

4.5.3: Delete in its entirety and substitute the following:

"4.5.3 Slow bend precrack charpy. The two specimens shall be taken from locations C1 and C2 on Figure 3. The test shall be conducted at a specimen temperature of -65 degree fahrenheit (^{O}F). The test shall be conducted in accordance with the procedure defined in the Appendix of this specification with the notches machined across the rear face of the specimen. Locations C3 and C4 on Figure 3 shall be used for replacement specimens for locations C1 and C2 respectively."

4.5.6: Delete in its entirety.

4.5.7: Delete the first two sentences and substitute the following:

"The unpainted warhead, after nose threading but prior to base threading and knurling, shall be mounted in a suitable test fixture. The warhead will then be subjected to the pressure requirements on the applicable drawing. Any warhead which fails to comply with the applicable reqirements shall be classed defective and removed from the lot."

Add "Figure 3" as follows:

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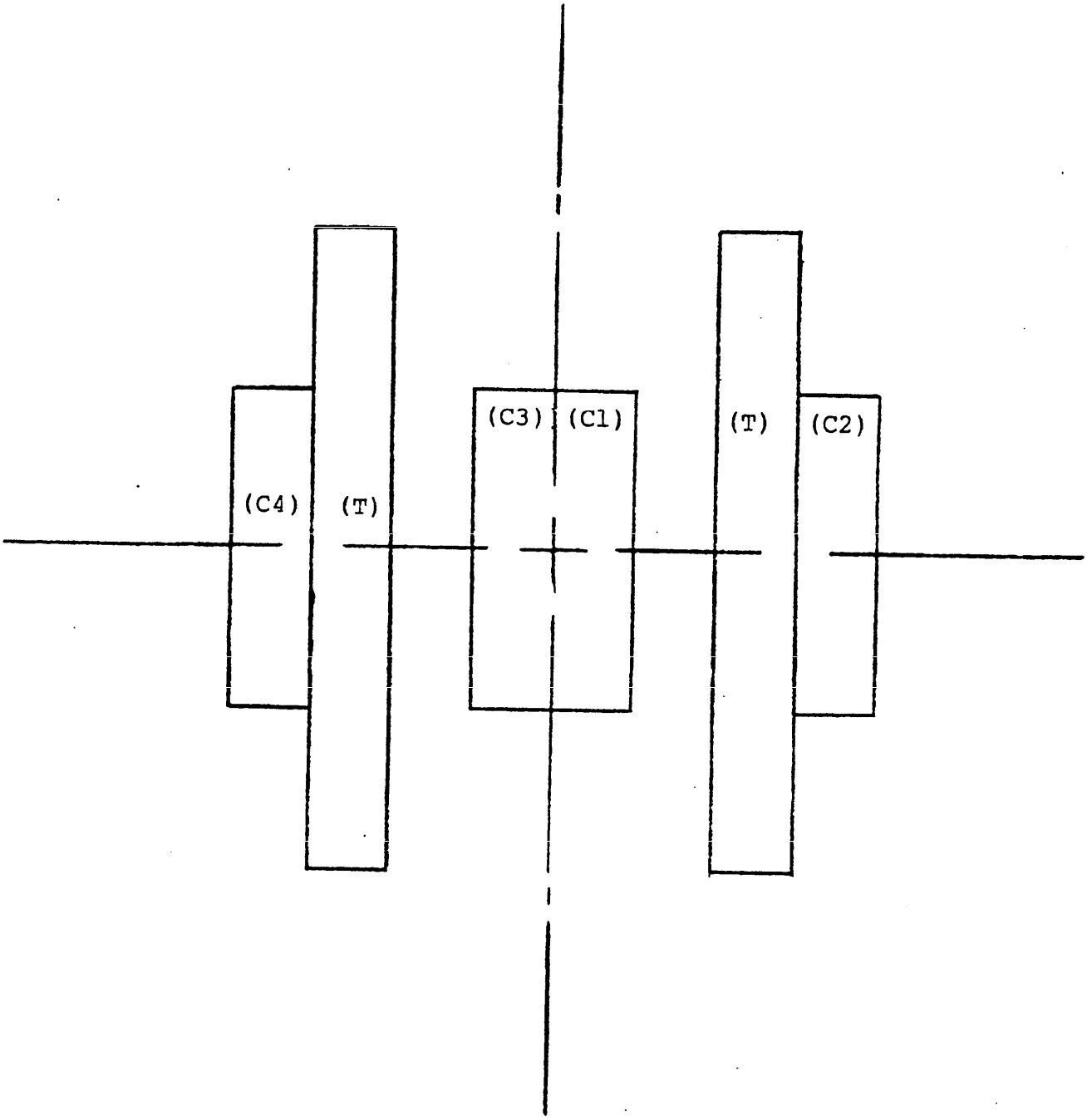


FIGURE 3. Approximate location of tension and charpy base specimens.

(T - Tensile, C - Charpy)

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4.5.8: Delete the first two sentences and substitute the following:

"The warheads shall be ultrasonically inspected after threading the nose but prior to threading and knurling the base using equipment and procedures in accordance with 4.5.8.1. With the exception of the forward threaded area, the warhead shall be inspected with ultrasonics throughout all of the material."

4.5.8.2: Delete "Calibration" and substitute "Verification".

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4.5.8.2: Add the following at the end of the paragraph:

"The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (see 4.5.8.4)."

4.5.8.4: Delete in its entirety and substitute the following:

"4.5.8.4 Operator qualifications. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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Add new paragraph 4.5.14 as follows:

"4.5.14 Depth of knurl. The depth of the radial groove in the knurl shall be checked using a dial indicating depth gage or approved equivalent."

5.1: Change "Level C" to "Level B".

Delete: "Section and Strippable Coating".

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* 6.3: Delete "ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801" and substitute "ARDEC, ATTN: AMSMC-QAR-I, Picatinny Arsenal, New Jersey 07806-5000".

6.5: Change "measured to gaged" to "measured or gaged".

* 6.7: Delete "ARRADCOM, ATTN: DRDAR-QAR-Q and DRDAR-LCU-CA, Dover, NJ 07801 and substitute "ARDEC, ATTN: AMSMC-QAR-Q and AMSMC-LCU-CA, Picatinny Arsenal, NJ 07806-5000".

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6.14: Delete in its entirety.

PAGES 29-35

* Delete "APPENDIX" and "FIGURE 1" in the their entirety and substitute the attached replacement pages.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes, (additions, modifications, corrections, deletions), from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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APPENDIX

A PRECRACKED CHARPY FRACTURE TOUGHNESS METHOD FOR HF-1 STEEL

10. SCOPE

10.1 Scope. This test method covers the determination of a fracture toughness, K_Q, of a Charpy-type specimen (see Fig. 1) containing a fatigue crack tested in slow bending. This test method will be applicable for HF-1 steel only.

This fracture toughness will be sensitive to changes in the plane-strain fracture toughness providing the strength of the specimen is determined primarily by crack propagation and not by plastic instability in the Charpy-type specimen.

20. APPLICABLE DOCUMENTS

20.1 ASTM Standards:

E-4 Practices for Load Verification of Testing Machines
E-8 Methods of Tension Testing of Metallic Materials
E-23 Methods for Notched Bar Impact Testing of Metallic Materials
E-139 Practice for Conducting Creep, Creep-Rupture, and Stress-Rupture Tests of Metallic Materials
E-399 Test Method for Plane-Strain Fracture Toughness of Metallic Materials
E-812 Crack Strength of Slow-Bend Precracked Charpy Specimens of High-Strength Metallic Materials

30. SUMMARY OF METHOD

30.1 Summary of method. This test method employs a Charpy specimen provided with a sharp notch terminating in a fatigue crack tested in three-point bending using fixtures that minimize the contribution of friction forces to the measured applied load. The maximum load in the test is recorded and the fracture toughness is determined from this value and the original dimensions of the specimen.

40. PRECAUTIONARY NOTES

40.1 Precautionary notes. When preparing specimens and testing for fracture toughness, it is important that certain precautions be kept in mind.

40.1.1 Toughness. The fracture toughness of some materials will be sensitive to the rate of loading and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The fracture toughness will be influenced by the testing temperature and calibration relations may change with testing temperature.

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50. DEFINITIONS

50.1 Fracture toughness, K_q. As determined by this method, the fracture toughness is defined as the nominal stress intensity in a notched and cracked bend specimen calculated on the basis of the length of the initial uncracked ligament and the maximum load sustained in the test.

50.2 The following symbols are used in this test method:

a - crack depth (machined notch plus fatigue crack) (in.)
B - specimen thickness (in.)
W - specimen width (depth) (in.)
f(a/W) - power series of ratio (a/W) from ASTM E-812
P_{max} - maximum load applied (lbf)
S - span between support rollers (in.)
K_f - maximum stress intensity factor during fatigue (KSI $\sqrt{in.}$)
K_q - fracture toughness obtained in the test (KSI $\sqrt{in.}$)
E - Young's modulus of elasticity (KSI)
T - test temperature (^{o}F)

60. APPARATUS

60.1 Tension-testing machine. Conform to the requirements of Practices E-4.

60.2 Bend-test fixture. The procedure involves testing of the specimen in three-point bending. The best test fixture should provide support rollers that are free to rotate and move apart slightly during loading (for example, see the Test Method E-399 bend test fixture for general principles of design). The nominal span between support rollers shall be 1.57 in. (between 1.50 in. and 1.65 in.). If the diameter of the support rollers is between 0.30 in. and 0.39 in. there will be negligible change in span during deformation of the specimen.

60.3 Temperature control. The testing temperatures shall be controlled by use of an integral temperature conditioning chamber as part of the testing apparatus. Any suitable method may be used to temperature condition the specimen providing the region including the notched and cracked section can be uniformly maintained within $\pm 2^{o}F$ of the desired test temperature.

60.4 Temperature measurement. The temperature of the specimen during testing shall be measured by means of two thermocouples attached on either side of the notched and cracked section. These thermocouples may be spot welded in place; however, the spot weld locations shall avoid the crack path. At the test temperature, the difference between the temperatures indicated by these two thermocouples shall not exceed $1^{o}F$ and the average of these two temperatures shall be within $\pm 2^{o}F$ of the desired test temperature.

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60.4.1 Thermocouples. Temperatures shall be measured with calibrated thermocouples used in conjunction with potentiometers or millivoltmeters. Such measurements are subject to various errors; reference should be made to ASTM E-139 for a discussion of errors.

60.4.2 Calibration. The temperature measuring apparatus shall be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that the accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS, AND PREPARATION

70.1 Dimensions. The precracked Charpy bend specimen is shown in Fig. 1. The crack length, a, (starter notch depth + fatigue crack length) is nominally equal to one half the specimen width, W, and shall lie within the range between 0.45 and 0.55 W.

70.1.1 Crack starter. The crack starter shall consist of a V-notch having an included angle from 45 to 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. is usually necessary in order to facilitate the initiation of a fatigue crack at a low-stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed:

70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

70.2.2 Method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three-point bending may be used to produce fatigue cracks.

Cantilever bending facilitates completely reversed loading (P_{min}/P_{max}= -1) which aids in starting and propagating the crack. For practical purposes, the K calibration for three-point bending may be used to estimate the stress-intensity factors for the cantilever bending.

NOTE: The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane that is normal to the notched face of the specimen.

70.2.3 Crack. The fatigue crack shall extend at least 0.04 in. beyond the tip of the crack starter notch. During at least the last 0.02 in. of fatigue cracking the following will apply:

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(1) The ratio of the maximum-stress intensity in fatigue to Young's modulus, $K_{f(max)}/E$ shall not exceed 0.002 $in.^{1/2}$.

(2) The maximum stress intensity in fatigue shall be less than 60% of the stress intensity from the test, $K_{f(max)}$ 0.6 K_q.

(3) The stress ratio P_{min}/P_{max} will fall between -1 and 0.1.

80. TEST PROCEDURE

80.1 Number of tests. Samples shall be selected as specified in the basic specification.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Fig. 1.

80.2.1 B and W. Measure the thickness, B, to the nearest 0.001 in. at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average value. Measure the depth W to the nearest 0.001 in. at two positions, one on each side of the crack starter notch and record the average value.

80.2.2 a. After fracture measure the initial notch plus fatigue crack length, a, to the nearest 0.5% at the following three positions: at the center of the crack front, a_3, and midway between the center and the intersection of the crack front with the specimen surfaces, a_2 and a_4. Use the average of these three measurements, $(a_2 + a_3 + a_4)/3$, as the crack length in the calculation of K_q (see 90.1). If the difference between any two of the crack length measurements exceeds 10% of the average, or if part of the crack front is closer to the machined notch root than 5% of the average, the specimen shall be discarded. Also, if the length of either surface trace of the crack, a_1 and a_5, is less than 80% of the average crack length, as defined above, the specimen shall be discarded. Record these five crack lengths.

80.3 Bend testing. Set up the bend-test fixture so that the line of action of the applied load shall pass midway between the support roll centers within 0.04 in. Measure the span (distance between the support-roll centers) within 1% and record this value as S. This value should fall between 1.50 in. and 1.65 in.

80.3.1 Location. Locate the specimen with the crack tip midway between the roll centers to within 0.040 in., and square to the roll axes within 2°. Check the parallelism between the top face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller shall be suitably adjusted.

80.3.2 Loading. Load the specimen at the rate between 700 lbf/minute and 3600 lbf/min. until the maximum load is reached, and record this value as P_{max}. Record the loading rate.

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NOTE: It is desirable to follow the load change carefully because specimens having low toughness will break abruptly and no slow decrease in load beyond the maximum will be observed.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 K_q. Calculate a tentative fracture toughness value in KSI $\sqrt{in.}$ from the measured value of P_{max} and the initial dimensions of the specimen as follows:

$$K = (P_{max}\ S/1000\ B\ W^{3/2})\ f(a/W)$$

90.2 Validity. Determine the validity of this tentative fracture toughness by checking the validity of the precracking (70.2.3), the shape of the crack front (80.2.2), and the bend test conditions (60.4, 80.3). If these requirements are met, this tentative value will be considered valid and will be reported as the fracture toughness, K_q, of the HF-1 steel.

90.3 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimen is given in Test Method E-399 and shall be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT

100.1 Report. The report shall include the following for each specimen tested:

100.1.1 All specimen dimensions (Section 80.2: B, W, a_1, a_2, a_3, a_4, a_5).

100.1.2 All test parameters (Section 80.3: S, Load Rate, P_{max}, temperature).

100.1.3 A check of all test validators:

100.1.3.1 Precracking.

A. $-1 \leq (P_{min}/P_{max}) \leq 0.1$

B. $K_{f(max)}/E > 0.002\ in.^{1/2}$

C. $K_{f(max)} < 0.6\ K_q$

100.1.3.2 Crack length.

A. $a_{fatigue} \geq 0.04\ in. + a_{notch}$

B. $.45W < a < .55W$

C. a_1 and $a_5 > 0.8\ a$

D. $a_i - a_j < 0.1\ a$; where $i,j = 2, 3, 4$

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100.1.3.3 Test conditions.

A. $1.5 \text{ in} < S < 1.65 \text{ in}$.

B. $700 \text{ lbf/minute} < \text{Load Rate} < 3600 \text{ lbf/min}$.

C. $T_{thermocouple} = T_{required} \pm 2^{o}F$

100.1.4 The fracture toughness, K_q, for all valid tests.

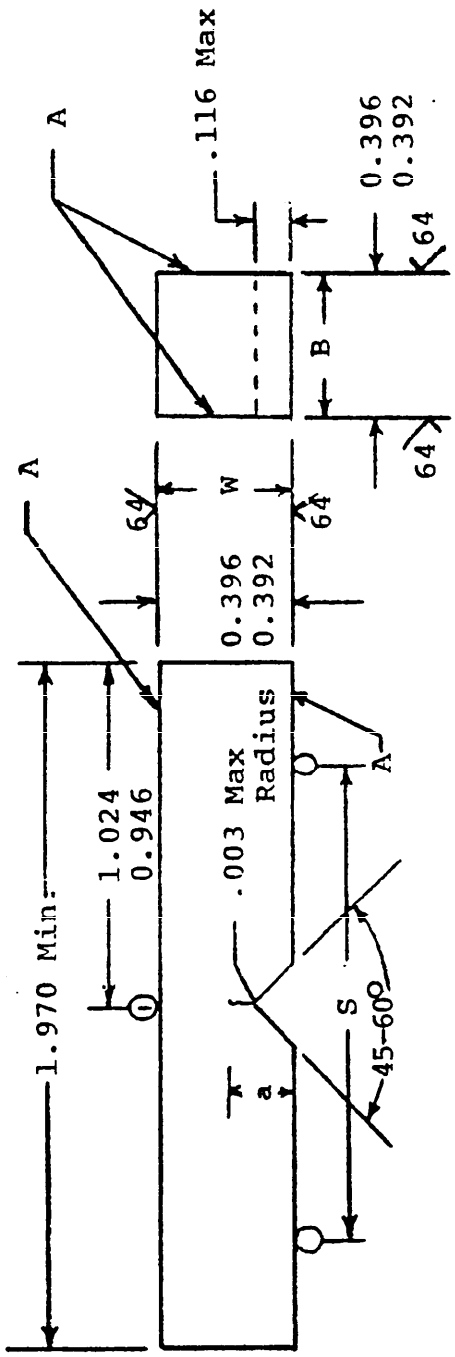
100.1.5 Crack plane orientation (ASTM E-399).

100.1.6 Fracture surface appearance (ASTM E-399).

100.1.7 0.2% offset tensile yield strength (ASTM E-8).

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APPENDIX



NOTE 1. All dimensions in inches

NOTE 2. "A" surfaces shall be perpendicular or parallel as applicable to within 0.001 In.TIR

NOTE 3. Crack starter shall be perpendicular to specimen length and thickness to within ± 2 degrees.

FIGURE 1 Dimensions of precracked charpy specimens.

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